Jargon Buster

ADSL2+ is a standard approved by the ITU (International Telecommunications Union) in January of 2003. It doubles the maximum frequency for downstream data transmission from 1.1 MHz to 2.2 MHz. If you're within 2 km of the CO, downstream data rates can be up to 24 Mbps. There's also an optional mode that doubles upstream bandwidth.

VDSL 2 was ratified by the ITU in May 2005. The standard delivers up to 100 Mbps both upstream and downstream, and works over copper up to 12,000 feet. VDSL 2 uses about 30 MHz of spectrum, versus 12 MHz in VDSL.

HDTV is short for "high-definition television." It's digital TV wherein a wide-screen picture is transmitted with many times more detail than current analogue television. HDTV requires five times the bandwidth of current-generation TV signals.

Modulation is the term for making one signal ride on top of another. The first signal we're talking about is the one that is of importance; the second signal is called the carrier. The carrier signal can travel better through the medium than the first signal alone. For example, in radio, we have Frequency Modulation (FM); the carrier signal is of the frequencies you hear about—105 MHz and so on. The actual data signal is used to modify this signal to a small extent, and it is recovered by the receiving equipment.

A **channel** is a frequency band in which a specific signal is transmitted. So when one refers to DMT splitting the spectrum into 247 channels, it's 247 separate frequency bands, and a signal on one band or channel will be independent of, and will not interfere with, a signal on a different band or channel.

The need (or the use) of fibre is what makes VDSL noteworthy as a technology. On the one hand, FTTN deployment may be difficult in some countries or areas; on the other hand, it's the fibre that enables VDSL to provide break-neck speeds over home telephone lines. And along the same lines, when it comes to ADSL2+ vs VDSL, it's a question of how far from the CO the deployment will be—ADSL is suitable for long-reach broadband, and VDSL will be viable if fibre itself is viable.

The Situation In India

VDSL in India? Far-fetched? We don't even have good ADSL yet! Anyway, we were wondering what the scene is like and what it will be, so we asked someone in the know—Jagbir Singh, group CTO, Infotel, Bharti Tele-Ventures Ltd.

First off, Mr Singh says DSL is the "global trend" and that it will remain the dominant technology for a long time. He also says things are "moving towards DSL, with around 44 million landline connections in the country and both the incumbent operators becoming aggressive." 44 million isn't too large a figure when you compare it with the penetration in the United States or even China. Still, it's there, whereas cable is virtually non-existent in rural India.

Broadband is definitely getting cheaper, with, for example, Airtel offering a wide range of affordable plans. However, Mr Singh says, "What we need today are policies that would replicate the mobile telephony growth success in the Internet arena. Unbundling of the local loop, peering for all at NIXI (National Internet eXchange of India), and Open Sky policy for DTH providers are a couple of steps that would allow better and cost-effective reach for any service provider to its target market segments."

Some explanation is warranted here. "Unbundling of the local loop" refers to privatisation of the last mile; currently, the government has a stronghold on what happens to the phone line between your home and the telephone exchange. We're not sure how long this will take.

The "Open Sky policy" refers to allowing DTH (Direct To Home) operators to have satellite uplinks from wherever they please; contrast that with the situation now, where they are only allowed to uplink from certain areas. Again, here, we don't know when the government will take positive steps in this direction.

"Peering over the NIXI" is when two ISPs agree to exchange traffic over the NIXI for free. An Internet Exchange, such as the NIXI, is a facility that allows ISPs to "meet" and exchange traffic. The purpose of establishing the NIXI was that Internet traffic that originates in India and has a destination in India should stay within the country; the advantage, of course, would be cost savings because of a reduction in reliance on international bandwidth. It would also increase speeds for obvious reasons—Indian traffic would not unnecessarily be routed all the way to, say, the UK.

We then asked about where the future of broadband in India lies—DSL, fibre, wireless, cable, what? Mr Singh is of the opinion that given the kind of content available and the needs of the Indian market as Bharti sees it, broadband connectivity on copper is "here to stay." Cable, according to Mr Singh, is still a long way from becoming a viable proposition—this, ostensibly because of the relative lack of an existing infrastructure.

Then, of course, we wondered about VDSL. We asked if Fibre To The Neighbourhood was feasible anytime soon. What Mr Singh said was a total surprise: he went beyond VDSL, and said that it was Fibre To The *Home* (FTTH) that holds the most potential as the technology for broadband services! Imagine an India with fibre running all the way to homes!

Of course, "rolling out FTTH quickly and in a cost-effective manner to cover rural as well as urban areas is a challenge," according to Mr Singh. "Deploying this technology means permissions from every residential block; there are administrative costs involved, and it takes time to carry fibre to every household.

"At present, ADSL delivers the best value proposition, both for service providers as well as customers."

So is it ADSL for now and FTTH for later? That's the way it would seem.

Some, however, are of the opinion that VDSL is the way to go for India. At the Convergence India 2005 International Exhibition and Conference, Moti Shulak, senior director, marketing, Infineon Technologies, spoke about the latest advancements in VDSL technology, and how they "fit perfectly to India's booming market."

In any case, we can conclude that the broadband situation is not as dismal as it might seem right now! Pricing, however, might make you gloomy again! ■

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